



Electric Multirotor Propulsion System Sizing for Performance Prediction and Design Optimization

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One of the more daunting tasks of designing a multirotor unmanned aerial vehicle (UAV) is the selection of a propulsion system that will provide desired performance. Rigorous methods for selecting these drive components, that is, the motors, propellers, and batteries for electric UAVs are not readily available. Currently, many UAV designs are based on legacy selections or limited and at times incorrect manufacturer data. These design methods are either simplistic or lacking in analysis and validation of component selection. Proper propulsion system design should address the mission requirements for which the vehicle is being designed. A proper design methodology is the best chance that the designer has to create a new vehicle that will be mission-capable. This paper attempts to satisfy the need for more thorough method of propulsion component selection. The paper is written also to document the popular online drive system validation tool due to numerous requests. This tool is one example implementation of the methodologies described by this paper.

Nomenclature

α	angle of attack
β	blade physical twist
B	number of blades per propeller
c	wing chord
c_R	c/R
C_b	discharge rating
C_D	drag coefficient
C_L	lift coefficient
C	energy capacity
dr	blade radial section
d_t	throttle command
dt	change in time
dL	incremental lift
dQ	incremental torque
η_E	ESC efficiency
η_M	motor efficiency
η_G	gearing efficiency
G	gear ratio
I_0	no load current drawn by 1 motor
I_a	current drawn by avionics
I_d	current drawn by all motors
I_m	current drawn by 1 motor

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I_p	current drawn by payload
I_t	current drawn total
K_v	motor speed constant
L	total blade lift
M_{tip}	blade tip Mach number
m_{avn}	avionics mass
m_b	battery mass
m_e	ESC mass
m_m	motor mass
m_p	total propeller mass
m_{pay}	payload mass
m_t	total vehicle mass
m_w	motor mass
N_m	number of motors
N_p	number of propellers
ω	rotations per minute
p_p	propeller pitch
p_{mat}	propeller composition
P_e	electrical power
P_m	mechanical power
Q	total blade torque
r	blade radial section location
r_R	r/R
R	rotor radius
R_b	battery internal resistance
R_e	ESC internal resistance, also R_{DSON}
R_m	motor internal resistance
R_w	wiring harness internal resistance
s	battery serial cells
S	wing span
θ	vehicle pitch w.r.t. local vertical
V_c	operational voltage per cell
V_s	voltage seen by motor coils
w	rotational speed rad/s

I. Introduction

The framework and algorithm presented is meant to be used to select and/or validate the selection of electric RC aerial vehicle drive components which will allow a vehicle to accomplish specific missions. As mentioned, one implementation of the methodology is available online, and is named the Electric Multirotor Sizing Tool (EMST). Missions are defined as hover, climb, or dash segments with associated durations. The method is capable of considering generic, standard and non-standard vehicle concepts including: airplanes, multirotors, single main rotor helicopters, coaxial configurations (motors and/or propellers), fixed or tilt wings/rotors/free wings, as well as generic combinations of the above.

A. Existing propulsion system design methods

The required fuel fraction method or RF method^{4,5,6} is used to size large, gas, or jet fueled aircraft such as manned helicopters. The method, when given a mission, provides a minimum engine size and thus attempts to size other components such as rotor blades and transmissions, as well as providing a minimum GTOW required after calculating other component weights. However, a similar method to size electric UAV drive components has not yet been found to be readily available.

Latorre⁷ documents a design process using BEMT to select motors, propellers, and batteries in an iterative fashion. This method should yield a reasonable result, but it does not optimize for any of the design

variables. Gur¹⁰ presents a multi disciplinary optimization (MDO) approach to design a propulsion system based on goals such as rate of climb and loiter time, and presents a useful modeling analysis of motors and batteries. Also presented are sensitivity analysis to certain propeller design elements. Lundström⁸ describes an automated approach to designing the propulsion system as well as the body for small airplane UAVs. Ampatis⁹ shows a design method using parameterized data from motors, batteries, and ESCs to study the effect of varying payload on design characteristics. Vehicle diameter, energy, motor length, and battery weight are described. The authors also consider the effects of varying the number of motors on the design. Also presented are thoughts on modeling of electrical propulsion systems. Bouabdallah¹¹ describes a method for iteratively designing a vehicle with a maximum mass and length to achieve a desired thrust-to-weight ratio. The method requires a database of actuator, battery, and airframe components to calculate the loop masses. The study presented in this paper, in contrast, attempts to parameterize drive components to rid of the need to use a database at design and optimization time.

Less rigorous methods in terms of optimality of UAV design also exist. For instance, eCalc^a has become one of the go-to tools for some UAV designers, especially hobbyists. Benito¹² describes a process of designing a multirotor using this tool. eCalc allows users to input specific lists of components and will provide a calculated flight time, in addition to other useful data. Similar to the validation algorithm described in this paper, it allows designers to check flight endurance and other characteristics of some types of common vehicles: heli/multirotors, planes, and ducted fans. eCalc requires that the specific drive components are provided to the tool. This is one of the major differences between eCalc and the validation algorithm described in this paper. That is, the EMST parameterizes and abstracts propulsion system characteristics to allow users to input only relevant drive system parameters instead of selecting specific components. In addition, the drive optimizer algorithm proposed in this paper is essentially the validator or eCalc in reverse, which the authors believe is at this point a unique capability. The algorithms presented here may be used in conjunction with or in lieu of existing tools such as this one either to validate each other, or provide starting points for design.

B. Multirotor electric propulsion system design considerations

There are multiple considerations that must be addressed when designing a multirotor UAV drive system. The scope of the design space may encompass some or all of the variables described in this section. There may be other considerations but the ones here are considered as primary design variables in that they greatly affect flight performance, mainly payload capacity and endurance.

1. Motor

Brushless DC (BLDC) motors are used for just about all of the vehicles described by this paper. These are generally preferred over older, brushed designs for their greater efficiency in converting electrical to mechanical energy. BLDC motor designs have two configurations, outrunner (OR) and inrunner (IR), which define the component of the motor that rotates and is attached to the output or actuation axle. IR motors spin an axle with magnets inside of array of windings arranged circularly around it. OR BLDC motors invert the IR design and wrap a bell of magnets around the winding array. Although IR motors allow for some tighter installations due to the body of the motor being static, the OR configuration allows those motors to produce more torque than their IR counterparts. This makes IR BLDC motors a common choice for many small (< 100 g) multirotor builds, and OR a good choice for anything larger. It is possible to use gearing (Parrot AR drone is a popular design) as well but due to the inherent complexity, many designs use direct drive BLDC OR motors. In a direct drive design, the propeller is directly attached to the motor axle. One benefit however of a geared design is the potential to hide a motor's axle mostly inside the motor body, making it more durable in the event of a crash. Of course, the gearing is now accepting the brunt of the crash energy, but this may be easier to replace than a motor axle.

Arguably the most important parameter of motor selection is the speed constant, or K_v , measured in RPM/V. It is the manufacturers indication of roughly how fast the motor will spin when unloaded per applied volt across its wires. This value must be properly matched to the selection of the propeller and battery, as well as chosen for resulting efficiency and lift capability. For the former, the motor should be able to handle the mechanical and torque electrical power loads applied by turning the propeller. Too high of a K_v or voltage, and the motor may be unable to handle the loads at high throttle, or may spin in a lower efficiency

^a<http://www.ecalc.ch/>

regime with too low of an RPM at low throttle. Too low of a K_v and the motor may have to spin too fast to generate sufficient thrust to fly. Generally a lower K_v motor that spins a larger and/or low-pitched propeller will be more efficient than a high K_v motor with a smaller and/or high-pitched propeller. One advantage of the latter however is higher performance for acrobatic flying.

2. Propeller

Propeller composition, radius, pitch, and number of blades must also be chosen to work properly with the chosen motor. The material from which the propeller is made might affect the efficiency of the propeller at different RPMs. This may occur for softer propellers due to flexure of the blades changing effective angle of attack at radial sections away from designed angles. This effect is described by Harrington.³ Increasing the propeller pitch and number of blades generally generates more thrust, but at a cost of efficiency and increased electrical and mechanical power requirements on the motor. Increasing the propeller's radius is generally more efficient, assuming the rest of the drive system is capable of handling the load. This is because the larger propeller, with all else being equal, may spin slower to generate the same lift. This allows the induced velocity to drop, thereby increasing propulsive efficiency.

3. Battery

This study considers common, current, hobby-grade batteries. Newer hobby-grade battery lithium polymer (LiPo) compositions are capable of specific energy of up to around 250 Wh/kg,¹⁵ about an order of magnitude lower than gunpowder, and two orders of magnitude lower than kerosene. Many COTS batteries are easily capable of discharging at currents greater than 100 A, some for extended periods of time. These batteries have all but replaced the previous NiCd and NiMH batteries which were used for hobby-grade vehicles in the past. Another composition known as lithium ion (LiIon), commonly used in consumer electronics such as laptops, has a specific energy 50% better than LiPo compositions, although maximum discharge rates are lower, which makes them better choices for more efficient vehicles.

Other compositions include lithium iron phosphate (LiFe or LFP), which are have a specific energy of around 100 Wh/kg. While they are heavier than other lithium compositions, they are considered to be more stable under discharge and when subjected to damage. Unlike LiPo batteries, they will not explode when punctured and their cells exposed to the air. These batteries are also reported to be able to withstand a greater number of discharges than other types mentioned above.

Aside from chemical composition, other main parameters must be considered when selecting a battery or batteries for the vehicle, although they are not necessarily independent. The battery's cell configuration and capacity must be considered. The battery must be chosen such that sufficient but not excessive voltage exists to power the propulsion system and avionics. LiFe cells have a maximum charged voltage of around 3.65 V/cell, whereas LiPo and LiIon cells are charged to 4.2 V/cell. Batteries are built by stacking cells in series (S) and parallel (P) to achieve endurance and discharge rate goals. Stacking cells in series increases the pack's voltage, while adding more cells in parallel increases its capacity. For example, a 4S1P LiPo battery has four cells in series and one in parallel, giving a fully charged voltage of 16.8 V, while the same configuration LiFe battery will be fully charged at 14.6 V. This value must correspond with the selection of motor K_v and propeller to achieve proper RPM and thrust values to lift the vehicle, along with any regulators that power other onboard electrical systems.

The capacity, normally measured in mAh, determines the energy storage of the battery. A 1500 mAh battery can provide 1.5 A of current for 1 hour if discharged to 100%. Batteries which are discharged to around 80% of their capacity per usage also tend to last longer than those which are discharged to a greater extent. The discharge rate, related to the so-called "C-rating" of a battery, is a manufacturer's indication of the discharge capability of the battery. A 40 C 1500 mAh battery in an operational condition is capable of sustaining 60 A discharge for some amount of time. The power requirements of the drive system must be taken into account when selecting the battery.

4. Electronic Speed Controller

At a high level, the electronic speed controller (ESC) takes a PWM control signal, generally 1 ms (off) to 2 ms (full), and converts it to AC current. The frequency of the switching, done by onboard FETs, spins the rotor on the BLDC. The main parameter when selecting ESCs to consider is the maximum rated

amperage. This must be sufficiently higher than the required drive current per motor such that the FETs do not overheat and fail. A secondary decision is whether or not to use ESCs with a battery eliminator circuit (BEC) onboard. These regulate main battery voltage down to usually 5 V to power other avionics. If this is not necessary, it is sometimes better to use optically isolated (OPTO) ESCs to save energy and heat, as the BECs tend to be inefficient and may cause unnecessary heating and energy loss.

II. Parameterization of drive components

A relationship was found between the masses and several key characteristics of all drive components considered by both the validation and optimization algorithms. These are necessary to abstract actual components into just the key characteristics to enable the algorithms to guess the masses of the components required to lift the vehicle. They are used in both validation and optimization algorithms, proposed in later sections.

A. Motor

The data shown here consider both IR and OR motors, the latter being a more popular choice for many multirotor configurations for their generally higher torque (per mass) and ease of installation. Figure 1 shows the trend of high K_v motors having low mass and vice versa, also studied by Gur,¹⁰ although with less motors, and with a broader focus on motor types, including heavy duty/high voltage motors. Both IR and OR motors are plotted in the figure.

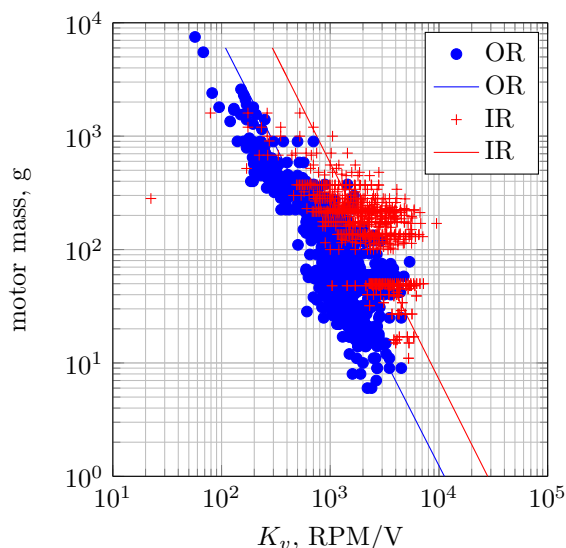


Figure 1. OR and IR K_v to mass relationship.

Figure 1 shows that in general, IR motors tend to be heavier per unit K_v than OR motors.

The trend in figure 1 generates eq. (1), providing a mapping between K_v and predicted motor mass m_m for both OR and IR motors.

$$m_m = 10^{p_1} K_v^{p_2}, g \quad (1)$$

where p_1 is 4.0499 and p_2 is -0.5329. For IR, p_1 is 4.4482 and p_2 is -0.5242. Of particular interest for this study are motors in the range of K_v and mass shown in Figure 2, as they are popular choices for the hobby-sized vehicles considered below.

Eq. (1) is modified to match some of the lower K_v motors available on the market, as the original fit does not match market survey data well in this region between 0 and around 500 K_v for hobby-grade motors. The modified equation for OR motors is seen in figure 2 as OR H. Around 991 OR and 696 IR BLDC motors were surveyed^b with incomplete entries removed.

^b<http://electrofly.free.fr/download.php?lng=en> accessed 03/10/2015

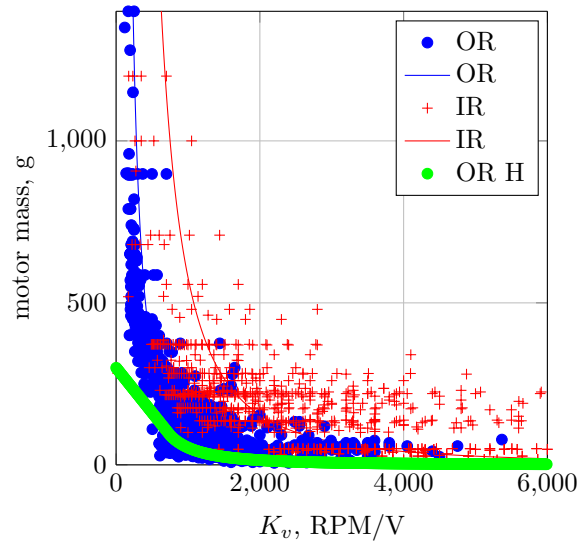


Figure 2. OR and IR K_v to mass relationship. OR H is the modified curve used for the validator and optimizer tools.

In addition to parametrizing the mass as a function of K_v , we are interested in the resistance of the motor, as generally, larger motors have larger electrical resistances. Gur¹⁰ shows that the internal resistance of the motor R_m is a function of K_v or mass.

B. ESC

Data from around 20 ESCs rated for up to 100 A are taken to study the relationship between ESC mass and current throughput capacity. These data are shown in figure 3. Eq. (2) is the function found that provides

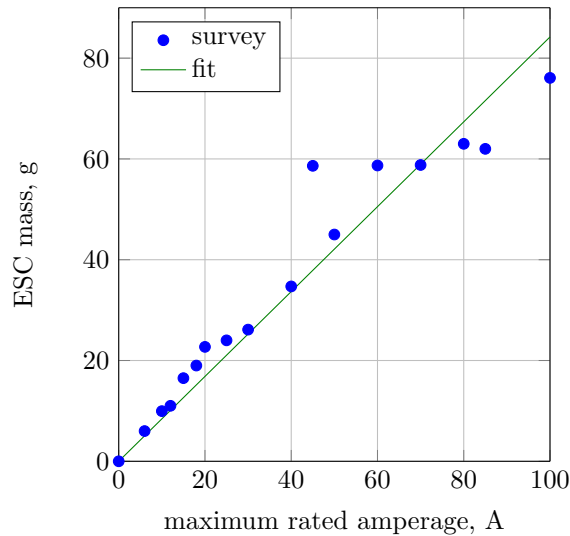


Figure 3. Masses of ESCs and their respective maximum amperage rating.

a mapping between an ESC's maximum rated amperage and its mass:

$$m_e = p_1 A_{max}, g \quad (2)$$

where p_1 is found to be 0.8421. In addition to ESC mass, it is useful to know the resistance of an ESC when it is operating. Ampatis⁹ shows that the resistance of an ESC R_e is between 9 and 45 $m\Omega$. This value is

$3R_{DSO\text{N}}$, where $R_{DSO\text{N}}$ is the resistance of one path from the input to the output of the ESC when the path is shorted.

Wiring, including all signal and power lines, has been found to weigh around 5% of the GTOW of all UAVRF multirotor vehicles. This fraction is used for all calculations by the tool.

$$m_w = 0.05m_t \quad (3)$$

Wiring resistance R_w is small and is scaled with the amperage required I_d .

C. Battery

A parameterization of battery mass and related capacity is also required for the algorithm to function. Data found are shown in figure 4.

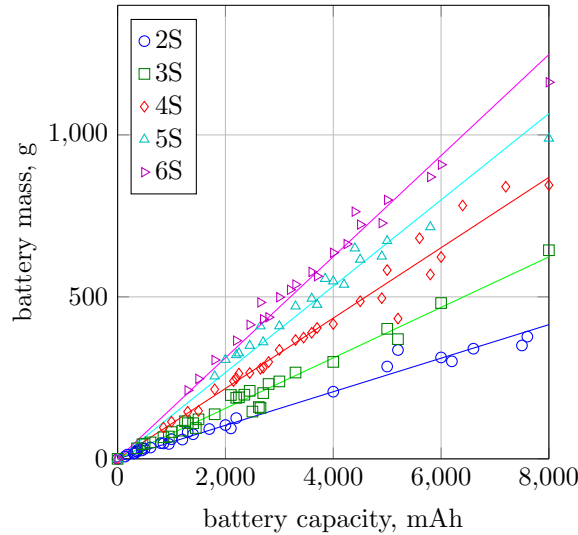


Figure 4. Battery masses for configurations between 2S and 6S. All batteries surveyed have 1P configuration.

Eq. (4) is the map between battery capacity and mass, assuming 1P configuration, calculated by considering approximately 30 common LiPo batteries^c. For the same chemistry, the mass of the battery is more or less linear with number of cells contained within it:

$$m_b = (p_1 s + p_2)C, g \quad (4)$$

where p_1 is 0.026373 and p_2 is 2.0499e-05. Gur¹⁰ also studies the relationship between battery capacity and mass.

As with the other electrical components, the battery's total internal resistance must be estimated. Ampatis⁹ shows the calculation of resistance of the battery R_b . This will also be used in the electrical model described in this paper.

D. Propeller

Propellers of varying compositions including nylon-plastic, carbon fiber, wood, and other plastic have also been characterized, providing a mapping between material, radius, and mass. Data from about 30 propellers are shown in figure 5.

The mass of the propeller is found by

$$m_p = p_1(2R)^2 + p_2 2R + p_3, g \quad (5)$$

where p_1 is 0.08884 and p_2 is 0 for wooden propellers, 0.05555 and 0.2216 for plastic, 0.1178 and -0.3887 for nylon reinforced plastic, and 0.1207 and -0.5122 for carbon fiber.

^c<http://www.hobbyking.com/hobbyking/store/index.asp> accessed 02/07/2015

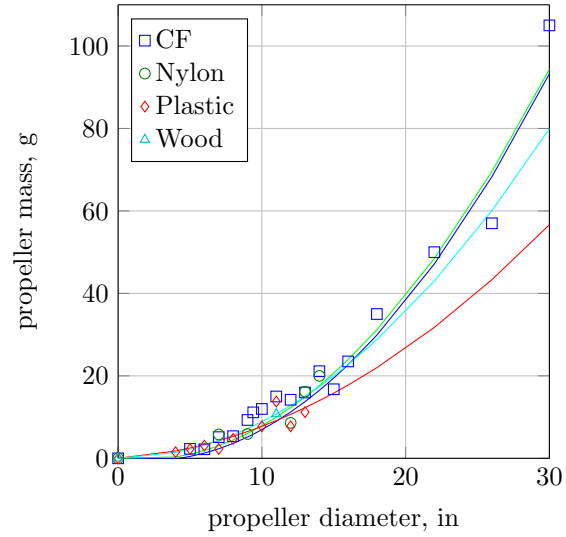


Figure 5. Masses of propellers of various diameters composed of four materials.

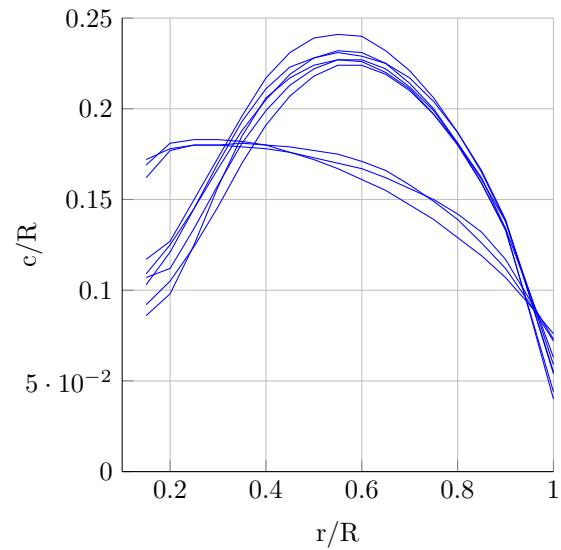


Figure 6. Non-dimensionalized propeller blade chord at radial section r/R . Data shown from GWS Slowfly and DirectDrive propellers.²

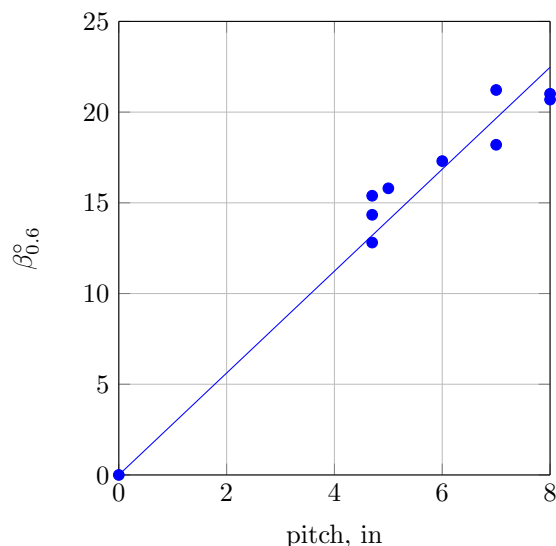


Figure 7. Twist of GWS blades at $r/R = 0.6$ compared to indicated blade pitch.

In addition to mass information, geometrical data are parameterized for propellers. This allows the use of a generic, averaged propeller if a specific geometry is not available or selected. Geometrical propeller data are taken from Brandt.² Specifically for this paper, we will consider GWS Slowfly and DirectDrive propellers, although other available data are also compiled and stored in separate, user selectable functions in the algorithm. This is necessary to feed the thrust calculation loop described below, as the lift per radial segment of the propeller is calculated based on these data.

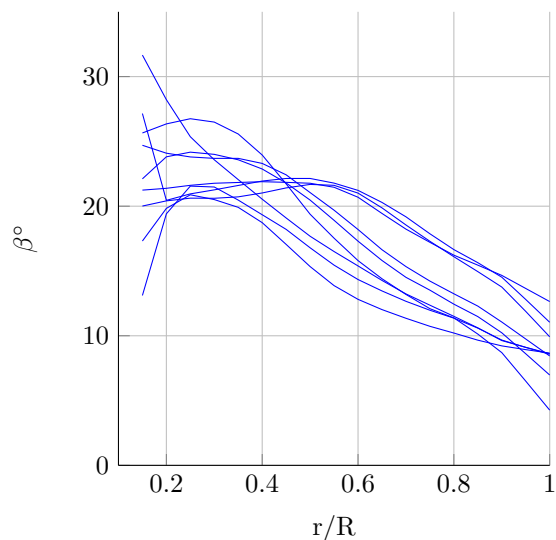


Figure 8. Blade twist at radial sections r/R for nine GWS Slowfly and DirectDrive propellers.²

Eq. 6 provides the physical twist angle β at the radial segment r/R for an existing set of GWS propellers and data from the above propellers are shown in figure 8.

$$\beta = p_1 r_R^3 + p_2 r_R^2 + p_3 r_R + p_4, \quad (6)$$

where, for this subset of propellers, p_1 is 30.322, p_2 is -64.731, p_3 is 23.008, p_4 is 20.558.

These data are averaged to provide a basis function for $\beta(r_R)$. This function is scaled by a function considering the pitch of the blade. The scaling is found by investigating the effect of pitch rating on the

actual twist $\beta(r_R)$ of the blade at $r/R = 0.6$. Pitch may be qualitatively defined as the linear, axial distance traveled by the propeller (at $r_R = 0.6$) should it be rotated by one revolution. This relationship between a pitch label and the physical twist β of a propeller may be seen in figure 7 for the GWS propellers.

p_p , β , and inflow velocity are calculated by the blade element and momentum theory (BEMT) algorithm and form the angle of attack of the blade section at r_R . If the flight condition includes a non-zero airspeed, the vertical component is added to V_0 to achieve the final angle of attack of the blade segment. Lastly, eq. 7 provides the chord at r/R for all nine GWS propellers described, also needed by the BEMT loop to calculate thrust. The distinction between the Slowfly and DirectDrive propellers is evident in the figure. For this subset of GWS propellers,

$$c_R = p_1 r_R^3 + p_2 r_R^2 + p_3 r_R + p_4 \quad (7)$$

where p_1 is -0.2872 p_2 is -0.1637, p_3 is 0.4551, p_4 is 0.05648.

E. Structural

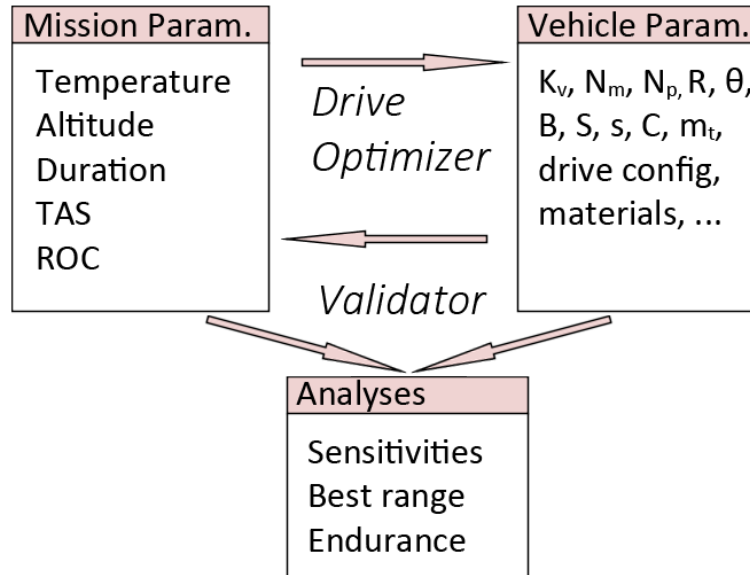
A survey of a few UAVRF vehicles was completed to parameterize the proportion of structural to GTOW. The algorithm assumes a very light-weight vehicle to have a structure of 8% of GTOW (GTQ Mini¹⁷), and a heavy-weight vehicle to have a ratio of 40% (GTQ2¹⁸). The ratio used for the discussion below is 19%. That is,

$$m_c = 0.19m_t \quad (8)$$

III. Overview of proposed design algorithm

The presented methodology is composed of two general components. The flight time calculator, or validator^d, is used when vehicle parameters are the inputs, and one is interested in determining capabilities of the vehicle. This might be used to validate, and hence the name, the selection of component parameters. If both the mission and vehicle design parameters are specified, the validator also allows for a sensitivity analysis of design parameters, as well as a best range and endurance analysis around the specifications. This is shown in the bottom part of figure 9.

Figure 9. EMST overview and workflow.



The drive optimizer algorithm is so called as it assists in mapping desired mission parameters to needed vehicle design parameters. That is, given a desired mission capability, such as a cruise for a certain duration, the drive optimizer will attempt to design a vehicle's propulsion system (as well as some chassis

^d<http://controls.ae.gatech.edu/dbershad/EMSTAirTimeCalculator.html>

parameters such as wingspan and chord, if desired) such that the vehicle's performance will be adequate to accomplish the mission. Both the validator and optimizer algorithms consider aspects of the following drive components: BLDC motors, electronic speed controllers (ESCs), propellers, and batteries. Non-drive components considered include payload, structure, and avionics, which are used for weight budget and power draw calculations.

A. EMST Validator

The validator may be used to substantiate the selection of components for a UAV design. This includes validation of the outputs of the optimizer algorithm described below. Given a propulsion system design, the algorithm allows a user to confirm that the vehicle will be capable of the mission for which it is designed. This is done in two ways. First, the electrical power requirements that the drive components will be subjected to in the mission flight condition are calculated so that relevant data of the selected components may be compared to manufacturer specified limits. This attempts to ensure that the selected components are capable of handling loads such as current, torque, etc. Second, the time endurance of the vehicle will also be calculated so that the user may confirm that the vehicle will be capable of completing the mission should the drive components handle the other requirements calculated.

The calculation method also provides a maximum range analysis. The analysis will provide the best airspeed for maximum range, and since the method is geared toward VTOL vehicles, the pitch angle from vertical required to attain it. In addition to this, the method provides a sensitivity analysis to design parameters, including motor K_v , propeller diameter and pitch, GTOW, battery capacity, altitude, number of motors, wingspan, etc. This helps identify key design parameters around the desired flight condition.

The validator algorithm performs all calculations based on the input of given or estimated characteristics of drive system components and their masses. Outputs of the method are described below and include hover time, design space sensitivities, drive system loading requirements in terms of RPM, torque, mechanical and electrical power, etc. Other data are also available as outputs of the validator. These are shown for an example vehicle below in table 3.

1. Validator hover time calculation

The validator algorithm first uses a loop to arrive at the linear throttle input d_t required to allow the vehicle to hover, climb, or dash as specified by the mission scenario inputs. The hover case is described first and cases with airspeed are described below. The throttle required to hover must be sufficient such that the total thrust calculated based on this throttle balances the weight force of the vehicle. The throttle required, along with the inputs described below, are passed to a BEMT calculator to calculate thrust generated by each propeller. The BEMT algorithm is also fed propeller characteristics, all of which are derived from models described in the parameterization and mission inputs. Throttle d_t is incremented until it is sufficient to accomplish the mission segment.

The vehicle's total mass must either be provided or estimated. The estimation of component weights is described in detail in the parameterization section. Eqns. 1, 2, 4, 5, and 8 are used to find the GTOW m_t . The m_t , number of motors N_m , motor K_v , number of propellers N_p , number of blades per propeller B , propeller radius R , propeller pitch p_p , and current validator loop throttle command d_t are passed to the BEMT algorithm to calculate the thrust generated by all propellers. Other inputs include results of eqns. 6, 7, the no-load current of the motor I_0 , current draw of other power sinks I_a and I_p , propeller material, battery configuration, composition, and qualitative quality, flight altitude and associated air density^e.

Together with the data above, a section lift and torque are calculated using the BEMT equations. For this study, the $C_L(\alpha)$ and $C_D(\alpha)$ are taken from the NACA 0015 profile.¹³ The entire blade section is integrated and which provides the total lift and torque applied by each propeller blade. This lift and torque is then scaled by the number of blades per propeller and the number of propellers on the vehicle.

The algorithm also allows for pitch-controlled propeller hubs. When the user selects such a hub, all hubs on the vehicle are assumed to be pitch-controlled. The throttle required sent to the BEMT by the outer loop is turned into a blade pitch command, subject to the maximum allowable pitch specified by the hub geometry. The blade planform is also selectable to be of a more traditional helicopter blade style, where $\beta(r_R)$ and $c(r_R)$ are assumed constant. Also in such a case, a governor may be enabled which controls the

^ecalculated using the ISA standard atmosphere

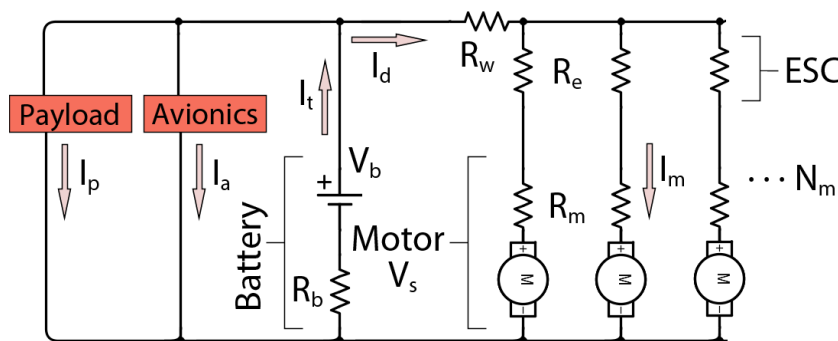
PWM command to the motor, thereby ensuring that the RPM at the head does not exceed a specified value. Otherwise, a headspeed may be specified for the BEMT calculations.

The algorithm has limited capacity for lifting bodies. Three types of lifting bodies are considered. First, the body of the vehicle has a selectable $C_L(\alpha)$ and $C_D(\alpha)$. If data or an estimate are available, this may be adjusted or ignored for the purposes of all calculations. The second and third types are wings. For these, the algorithm accepts $C_L(\alpha)$ and $C_D(\alpha)$. Wings may be either fixed to the body via an incidence angle, or free/actuated and controlled to a specified α .

2. Electrical modeling

The algorithm calculates the power and current required to hover. It does so by summing the torque components on all of the propeller blades on the vehicle. Standard equations of power are applied to all components shown in figure 10.

Figure 10. EMST propulsion system electrical model.



Assuming a steady hover, the mean voltage seen by the motor coils V_s of one motor is

$$V_s = d_t s V_c - I_d (R_b + R_w + R_e + R_m) \quad (9)$$

Drive current I_d is the sum of current for all motors, and I_m includes the no load current I_0 for each motor. It is also important to account for the change in internal battery resistance R_b as current demand changes. Current pulled through this resistor will generate heat and this power must be accounted for in the analysis. The effect is described by Stepaniak.²⁰ For all analyses, ESC efficiency η_E is assumed to be 97%^f. Wiring resistance R_w is taken from standard copper wire, assumed to be of a particular gauge and length, scaled with the current requirement I_d . Internal resistance of the battery R_b is scaled with the qualitative condition of the battery specified by the user and as described by Ampatis.⁹

Motor RPM is calculated based on the voltage seen by the motor coils. This is sent to the BEMT for the torque calculation.

$$RPM = K_v V_s / G \quad (10)$$

This no-load RPM is for directly driven or geared drive trains. It is important to note that the actual RPM will be lower as load is applied to the motor. The reduction in RPM depends on amount of torque applied, and generally for the hobby class of propellers discussed here, RPM has been found to decay at a rate of approximately 15-20% per blade added to the propeller. Current increases by the same amount given the same d_t . Gear efficiency η_G is also considered at this point, i.g. about 3% increase in power required for a spur gear. Helicopter governed head speeds may also be specified as applicable. The motor current I_m is calculated using the BEMT output of torque:

$$I_d = N_m (Qw / (\eta_M V_s) + I_0) / \eta_E \quad (11)$$

Drive, payload, and avionics currents are summed to arrive at the total current requirement in the given condition.

$$I_t = I_d + I_p + I_a \quad (12)$$

^fScorpion Calc v3.37: <http://www.scorpionsystem.com/downloads/> accessed 01/13/2015

This should include other power sinks such as tail rotors for helicopters (around 5%¹), coaxial propeller losses (around 15%¹ depending on separation), etc. To calculate the flight time, the algorithm compares the requirement I_t and the battery capacity. A discharge percentage less than 100% is generally used and may be specified by the user to elongate the life of the battery.

3. *Flight time calculation, non-hover*

Other forces are also considered for non-hover cases such as climbing or cruising. These forces include lift and drag of fuselage and lifting body components. Starting at a vertical thrust axis aligned with gravity, the algorithm uses an iterative loop to calculate the angle required to achieve a desired airspeed. Constraints are set so that the vehicle does not lose altitude during the cruise. A climb rate may also be specified. The method considers lifting bodies and control, their incidence angles, as well as controlled wings that keep a constant angle of attack with the free stream. For each α , the vehicle's total L and D is calculated. The loop quits when the total lift is greater or equal to the weight force of the vehicle.

In the outer, throttle required loop, the angle calculated by the above loop is used to add axial flow into the propellers. V_∞ , the axial component of the TAS, is added to the V_0 component in the BEMT calculator.

4. *Range calculation*

The validator algorithm attempts to calculate the vehicle's time endurance for a range of airspeeds. Using these with the angular iteration above, a maximum range is available for each condition, providing roughly "optimal" airspeeds for maximum range and maximum endurance. Of course, this is range in the wind reference frame, so any wind would have to be taken into account for planning purposes.

5. *Sensitivity analyses*

The validator algorithm calculates hover time sensitivities to design parameters. The sensitivities allow a user to incrementally move around the design space and evaluate the effect of the parameter shift in question on the hover time capability of the vehicle. These parameters include commonly changed ones such as motor K_v , propeller pitch and radius, GTOW, battery capacity, and hover altitude. Hover altitudes use both MSL and AGL, the latter for hover in ground effect (HIGE).

For each sensitivity, a linearisation of values, for example, dt/dK_v , are derived around the set point, where dt is the change in hover time and dK_v is the desired K_v increment. The validator is rerun changing $K_v \pm dK_v$. The two resultant values are then used to calculate a linear dt/dK_v at the set point. Of course, as the increment becomes larger, the linear assumption deteriorates, but should be sufficient for most conditions with "small" increments. For battery capacity and propeller radius, the algorithm will calculate additional masses resulting from adding/subtracting increments of both parameters. Sensitivity analyses for several parameters are shown below for two vehicles at the UAVRF.

6. *LiPo cold degradation*

Battery performance is affected by the temperature in which the battery is operating. Operation in cold or hot environments drastically affect^{15,16} the performance of LiPo and LiIon batteries. Some manufacturers claim that the optimal temperature for LiPo operation is above 68-70 °F^g. LiIon data suggests 77-100 °F is even more ideal.^h While extensive thermal range testing of batteries has not been studied by the authors, they were able to test several batteries in around -20 °F temperatures. Using the performance data that were collected during these flights, and anecdotal data acquired on online user groups, a simple performance degradation factor minimized at 70 °F is created as a crude way to account for extreme temperatures. Both the capacity and voltage of the battery are reduced exponentially at different rates far away from this optimal temperature. These trends may be observed in the LiIon data sheet.

^ghttp://batteryuniversity.com/learn/article/discharging_at_high_and_low_temperatures accessed 11/01/2015

^h<http://industrial.panasonic.com/lecs/www-data/pdf2/ACA4000/ACA4000CE417.pdf> accessed 11/01/2015

7. Ground effect

A ground effect correction¹ is applied to all flight calculations. The power required to hover is scaled by the HIGE correction described, which is a function of the AGL altitude of the vehicle, which is assumed to be the same as separation of the propeller disc and the ground.

8. BEMT inflow model

The BEMT model runs in two modes; one considering inflow and one without. Inflow velocity is calculated by iterative means. For general, hobby-sized vehicles in the range of 0-2 kg, the flight times only vary about 1-2% between the two BEMT solutions whereas the computation time may vary up to +20% for each loop. As it is not readily clear which is more accurate considering other sources of error, it is recommended that the inflow model not be considered for most calculations unless forward flight is required.

9. Major error sources

One of the greater sources of error for this methodology implementation is the propeller airfoil assumption used when a specific propeller's geometry is not used. The best case is of course using the geometry of the propeller to be flown. For example, in all analyses described in this paper, the NACA 0015 airfoil is for all BEMT calculations. This airfoil, of course, is generally a poor assumption, especially for propellers with higher pitch values. This is due to the zero camber of the symmetric NACA 0015 as compared to the non-zero cambers of most propellers available in the hobby market. Bohorquez¹⁴ shows that C_p/C_t of a propeller can vary greatly when a non-zero camber is used as opposed to when the airfoil is symmetric. This may greatly change the thrust and power calculations produced by the BEMT.

B. Drive optimizer

The drive optimizer is a proposed method to design and optimize the propulsion system of generic single/multirotor vehicles. The output of the methodology is generally a propulsion system which will allow a vehicle to accomplish a specific mission. Depending on the objective of the optimizer, the vehicle is sized for either lightest weight, smallest wing, highest efficiency, highest climb rates, or other performance objectives. The propulsion system which is sized consists of a motor, propeller, battery, and ESC. The work thus far focuses on electric vehicles, although the process is valid for other energy sources with some modifications. The methodology is developed to allow a designer to generate a vehicle that will be capable of completing a specific mission.

As seen in figure 11, until triggering a termination condition, the algorithm will loop through potential propulsion and other parameters (e.g., K_v , p_p , S , etc.) to find acceptable solutions to the mission problem requirements. The algorithm will then select the best configuration depending on desired optimization parameters. For example, in the default mode (seen in the figure), if the lightest possible vehicle is desired, then the configuration with the lightest GTOW will be selected. The algorithm estimates the masses of all subsystems per iteration using the characterizations of the drive, structural, and electrical components discussed in previous sections. This is done for all components for which mass is not provided, i.e., for components that are not specifically selected by the user. For each iteration, the sum of all drive and structural components is used arrive at a total gross takeoff weight (GTOW).

To save time, the loop termination condition will be set to exit the loop when a feasible configuration is found while increasing incrementally the allowable, available mass. This ensures the lightest configuration is found, within the range of the mass increment, and of course, other calculation errors.

1. Feasibility of optimizer solutions

For all calculations, the method abstracts the drive components; that is, it does not need to know the specific motor, propeller, ESC, and battery used, and instead relies on their parameters. To illustrate this, consider a motor. The method does not need to select a specific motor from a database for calculation of capabilities and instead uses standard motor parameters (ie, K_v , I_0 , etc). However, because of this abstraction, motors, propellers, ESCs, and batteries of either the validator or the optimizer's resulting configuration need to be checked before implementation. For instance, the motors are assumed to be able to handle the required current for hover or the specified flight condition for a particular mission. The manufacturer's specifications

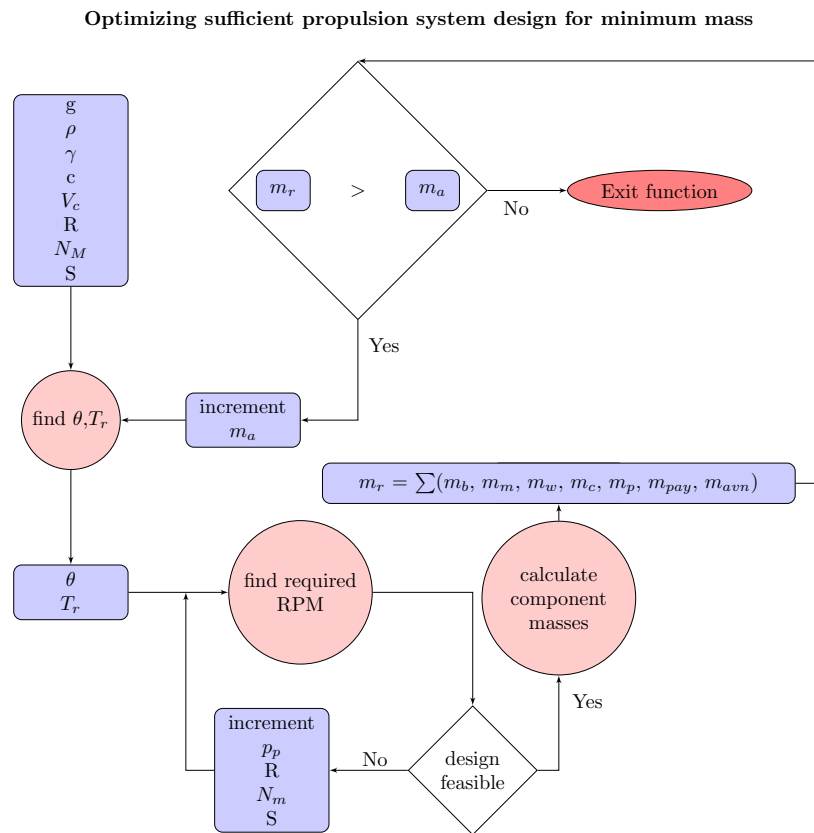


Figure 11. Optimizer overall flowchart. This particular optimizer is set to find the lightest possible vehicle to accomplish the specified mission segment.

of maximum power of motors must be confirmed to be higher than the power required for the model. In addition to this, propellers and structural components are assumed to be able to handle dynamic and static forces experienced in flight without yielding.

IV. Results

A. Validation of validator algorithm

A survey was done in an attempt to validate the modules (BEMT, component parameterization functions, etc.) that are used in both the validator and optimizer. The survey compares reported maximum hover or flight times and those calculated by the validator. Table 1 shows results from vehicles that were tested. For each vehicle to be tested, the vehicle's parts characteristics must be available. All vehicles in the table have corresponding characteristics including, at a minimum, motor K_v and number of motors, propeller diameter and pitch, battery configuration and capacity, and GTOW, all available from each respective source. Of course, the nature of the survey brings with its data a certain anecdotal aspect, although with a large enough sample size, results should be valuable. Several of the vehicles surveyed are part of the UAVRF fleet and the authors have averaged endurance data from hundreds of flights. Of the vehicles surveyed, the mean absolute error between reported and validator-calculated flight times is 5.7%.

1. Application of the EMST algorithm

The optimizer and validator tools were used in the design of several vehicles for the UAVRF. This discussion will focus on one in particular. The GTQ-Mini¹⁷ "Frobenius" is the winner of the 2015 AHS MAV Challenge held in Virginia. GTQ Mini was built specifically for the competition by the Georgia Tech Aerial Robotics (GTAR) team. The vehicle was to maneuver in a GPS-denied environment with no external navigation aides and was to comply with a 500 g maximum GTOW. GTAR decided to employ a vision algorithm which required a heavy, i7 processor to deploy successfully. The computer used requires an average 3 A draw at around 16 V (4S). To select the drive system, the drive optimizer was used. The results of the optimizer are seen in table 2. R was constrained to 2.5 inches due to physical size limitations imposed by the competition and design. The other inputs of relevance are a 10 minute desired hover endurance and a maximum weight of 500 g.

The Multistar 1704, 1900 K_v motors fit the weight and power budget and although they do not match closely with the optimizer output, they at least come the closest of the motors which were available to GTAR at the time. Note the increase in I_d when switching from 1383 to 1900 K_v . These motors were tested to ensure that they are an appropriate choice. Because the EMST tool was largely untested at the time of design of this vehicle, combinations of 3S and 4S battery, 5030 (shorthand for 5 in diameter, 3 in pitch), 5030x3 (3-bladed), and 6030 propeller configurations were tested. The combination with the best balance of thrust, flight time, and motor temperature was 4S 5030 for this motor. In contrast to what is shown by Mulgaonkar,¹⁹ this vehicle draws about 65% of the total power for propulsion in hover due to the power-hungry i7 computer and small motors.

Table 3 shows the validator outputs for the vehicle. The 850 mAh batteries keep the vehicle in the air for about 6.5 minutes and in weight for the competition, although the 1300 mAh battery used for testing keeps it in the air for about 9-10 minutes, depending on allowed level of battery discharge.

A maximum range analysis is also performed for the vehicle assuming perfect 850 mAh batteries and HOGE (hover out of ground effect), shown in table 7. These data have not been validated in flight test. Table 3 also predicts the propeller RPM during hover to within the resolution of the tachometer used to measure the RPM during an actual flight.

As seen in table 3, the validator method predicts an I_d of 4.49 A for hover. When added to the i7 I_p of about 3 A during heavy calculations, the total current required to hover I_t becomes about 7.5 A. This matches well to what is observed when a current meter was used during a hover flight. Table 4 shows other compared values from the actual vehicle.

Note the algorithm's estimates for all components but payload. In this case, the payload is the i7 computer and related avionics, which has an installed weight of about 175 g including a RAM module, USB serial devices, and WiFi module with antennae. Table 4 shows the estimates as compared to actual measured weights used on the competition vehicle. Other than that and the structural component, the predictions match closely to what is observed. The structural component is off because eq. 8 assumes that the structure

Table 1. Flight time calculation validation survey

Type	Vehicle	Reported endurance min	Calculated endurance min	Δ %
QR	GTQ Mini	5.5 ^a	5.8	-5.5
QR	Quadshot (hover)	7 ^b	9.2	-31.4
QR	HS 250	7.5 ^a	6.9	8.0
COAX	Helibaby	8 ^c	7.8	2.5
QR	BlckBd QR	8.5 ^a	8.4	1.2
QR	Blade Nano QX	10 ^d	9.5	5
QR	KMEL500	11 ^e	11.6	-5.5
QR	DIYD QR 1 (Gstv)	12 ^f	12.1	-0.8
SMR	T-REX 600e	13 ^g	12.8	1.5
OR	3DR X8	15 ^h	14.9	0.7
BWB	Quadshot (cruise)	15 ⁱ	14.9	0.7
QR	DJI Inspire	15 ^j	14.8	1.3
HR	MB Hexa	15.5 ^a	16.7	-7.7
QR	Hummingbird	18.3 ^k	18.9	-3.3
QR	3DR Iris+	19 ^l	18.2	4.2
QR	DJI Phantom 1	25 ^m	25.5	-2
QR	QR 4	73.5 ⁿ	72.3	1.6
QR	QR 3	87 ⁿ	73.5	15.5
QR	QR 1	97.1 ^o	101.7	-4.7
QR	QR 5	109.7 ⁿ	107.5	2
QR	QR 2	129.2 ⁿ	102.5	20.7
mean err, %:				0.4
mean abs err, %:				5.4

^a<http://www.uavrf.gatech.edu/> accessed 01/22/2015^bhttp://thequadshot.com/w/index.php?title=Quadshot_Technical_Specifications accessed 02/19/2015^chttp://www.tech-sov.com/news_13.htm accessed 02/22/2015^d<http://www.bladehelis.com/Products/Default.aspx?ProdID=BLH7600> accessed 02/18/2015^e<http://kmelrobotics.com/> accessed 02/19/2015^fhttp://diydrone.com/group/arducopterusergroup/forum/topics/new-vehicle-design-tool?xg_source=activity accessed 09/17/2015^g<http://rc.runryder.com/helicopter/t669735p1/> accessed 02/19/2015^h<http://3drobotics.com/x8/> accessed 02/19/2015ⁱhttp://thequadshot.com/w/index.php?title=Quadshot_Technical_Specifications accessed 02/19/2015^j<https://www.dji.com/product/inspire-1> accessed 06/06/2015^k<http://www.ascotec.de/en/uav-uas-drone-products/ascotec-hummingbird/> accessed 03/14/2015^l<https://store.3drobotics.com/products/iris> accessed 02/19/2015^m<http://www.dji.com/product/phantom> accessed 07/22/2015ⁿ<http://www.rcgroups.com/forums/showthread.php?t=1880665> accessed 01/22/2015^o<http://diydrone.com/profiles/blogs/my-97minute-06second-record-quadcopter-flight> accessed 01/22/2015

Table 2. Drive optimizer outputs for GTQ Mini design.

Parameter	Value
s	4
K_v , RPM/V	1383
C, mAh	936
R, in	2.5
p_p , in	3.0
Ω_{hover} , RPM	11600
I_d , A	3.52
m_t , g	496.3
m_b , g	104.7
m_m , g	116.1
m_e , g	37.9
m_p , g	11.6
m_{pay} , g	165
m_{avn} , g	20
m_w , g	13.5

will be less efficient in terms of weight than what was produced in order to be in weight for the competition.

A sensitivity analysis is also performed at the design point. These parameters are varied with all other parameters held constant. It is possible to run this with an MDO wrapper to provide an indication of where design efforts should be concentrated. Table 5 presents the inputs taken by the algorithm, whose outputs are shown in table 6.

According to the algorithm, the vehicle, around the setpoint described, would benefit most in terms of flight endurance by increasing the battery capacity. Increasing any other value would reduce the flight time and vice versa. Although this has not been rigorously validated and is, of course, a linearization, the change of battery from 850 to 1300 mAh increases flight time by about four minutes, roughly equivalent to the algorithm's prediction of 0.84 min/100 mAh. That is,

$$(0.84 \text{ min}/100 \text{ mAh})(1300 - 850 \text{ mAh}) = +3.78 \text{ min predicted} \quad (13)$$

which is approximately correlates to what is observed: approximately a four minute increase in endurance. Physically changing the other parameters has not yet been tested. Flights are terminated when any battery cell reaches 3.6 V (loaded) to extend battery life.

V. Conclusion

A method for design and validation of design for a generic UAV electric drive system has been presented. The method is based on designing a vehicle for a particular mission. The method considers geometric characteristics of propellers, aerodynamic, and electrical aspects of the system to be designed. Two algorithms are discussed to map bidirectionally between a mission profile and a set of UAV drive system's parameters. While flavors of the mission validation algorithm exist, the algorithm presented is more general than most in terms of vehicle configurations. Also, a sensitivity and range analysis are provided. The sensitivity analysis allows a designer to study the effect of nudging the design space on flight performance. The range analysis provides insight into the maximum range and best calculated flight speed/attitude to achieve it. Other than the work presented here, algorithms for the reverse direction, from mission to components, are not readily available.

Other optimization design methods, described above, require a database of components to calculate performance characteristics of a vehicle. The presented algorithm abstracts specific component databases into relevant parameters, eliminating the need for database upkeep and data mining. In addition to this, the algorithm is directly set up for an MDO approach to enhance the optimizer's performance. The algorithm

Table 3. Validator general system outputs for GTQ Mini at hover

Battery C required	6	C	Component total: motors	64	g
Throttle / collective required	41.25	%	ESCs	62.29	g
Drive current required	4.49	A	Batteries (drive)	95.39	g
Motor electric power / motor	17.93	W	Payload	0	g
Motor mechanical power / motor	12.91	W	Structural	95.41	g
Efficiency: specific thrust / motor	7.01	g/W	Wiring	24.95	g
Battery temperature coef. V	100	%	Avionics	50	g
Battery temperature coef. C	100	%	Propellers	4.68	g
Vehicle pitch angle	0	°	GTOW	499	g
Propeller speed	11599.5	RPM		80%, 100% dt	
Motor speed	11599.5	RPM	Thrust/weight	3.79, 5.92	-
Tip speed	77.13	m/s	Specific thrust	2.64, 2.83	g/W
Tip Mach number	0.23	Mach	Current/motor	2.80, 4.09	A
Motor torque / motor	0.00352	Nm	Motor electric power	44.73, 65.26	W
Disc loading	9.85	kg/m ²			
Power loading	28.07	kg/kW			
Max governor	0	%			
Blade pitch	3	°,in			
Mixed flight time	4.08	min	Hover time	6.81	min

then may easily be used to find propulsion systems with high loiter times for endurance or high thrust-to-weight ratio for adeptness in acrobatics and high rate of climb. Allowing this departure from component databases is the key to a truly optimized drive system, and potentially a faster and less limited path to it. That is, for the former, the algorithm does not need to cycle through the data from thousands of motors in the database to find an appropriate one. For the latter point, a database-based optimizer is limited to a finite number of solutions, directly proportional to the number of components stored. The presented algorithm should thus be able to traverse a greater design parameter space than any that relies on a database.

The new design algorithms are applied to several vehicles, and one such application has been described. The GTQ Mini is the winner and only finisher of the autonomous category in the 2015 AHS MAV Challenge. The main outcomes related to this work were described in the results section. The algorithm was able to accurately predict required RPM, throttle, drive current, flight time, component weights, and battery capacity sensitivity for the vehicle. Other data have been outputted as well but have not yet been validated. The system has also been run against a multitude of existing vehicles and found to predict their reported flight times to within 5% on average. Vehicles which were studied include helicopters, multirotors, and a blended wing-body VTOL vehicle.

To allow these algorithms to function, a parameterization of motors, ESCs, batteries, and propellers has been presented. A trend between motor's speed constant and its mass was found. This allows for the prediction of a motor's mass based on knowledge of K_v . Similarly, prediction of a battery's mass is possible based on knowledge of its composition and configuration. The drive current required allows for predicting the mass of the ESC and wiring, and the diameter and material of a propeller blade is an indicator of its mass.

A simple electric propulsion system model has been presented. One area for improvement is in the motor model. More work needs to be done to model the RPM performance of a loaded motor, especially when B is greater than 2. Also, current limitations of motors should be parameterized to throw away impossible configurations when optimizing the drive system. Another such area is the battery model which does not include V_c sag dynamics. When a LiPo battery is stressed, the cell voltage may drop abruptly, especially under 3.5 V for the batteriesⁱ used by the UAVRF.

The aerodynamics model of the propellers would benefit from knowledge of a more accurate airfoil of

ⁱVarious 3S Thunderpower, Zippy, and Turnigy Nano batteries between 20-90 C

Table 4. Validator predictions from table 3 compared to actual measurements for GTQ Mini in HOGE.

Parameter	Predicted	Measured
RPM	11600	11560 mean
I_t , A	7.49	7.8 mean
Motors, g	64	68
ESC, g	62.29	38
Battery, g	95.39	96
Structural, g	95.41	41
Wiring, g	24.95	20
Avionics, g	50	47
Propellers, g	4.68	6.8
Payload, g	-	175

Table 5. Sensitivity analysis inputs for GTQ Mini at hover.

Parameter	$+/- \Delta$
K_v , RPM/V	100
Prop diameter, in	0.5
Prop pitch, in	0.5
GTOW, g	50
Capacity, mAh	100
Altitude, m	3000
TAS range, m/s	0-15.24

the propellers modeled instead of the NACA 0015 used, which is symmetric. The ground effect correction applied is only valid in a hover case and needs to be updated to include downwash from wings and forward flight effects. Tip losses are also currently not modeled.

Acknowledgments

The authors would like to thank the members of the UAVRF and the RC flight community for their help with this project.

Table 6. Sensitivity analysis outputs for GTQ Mini at hover

Sensitivity parameter	Specific sensitivity		Total sensitivity	
dt/dK_v	0.0018	min/K_v	-0.18	$min/100K_v$
dt/dD	-0.66	min/in	-0.33	$min/0.5in$
dt/dP	-1.29	min/in	-0.64	$min/0.5in$
dt/dm	-0.01	min/g	-0.33	$min/50g$
dt/dC	0.01	min/mAh	0.84	$min/100mAh$
$dt/dAlt$	0.000393	min/m	-1.18	$min/3000m$

Table 7. Maximum range analysis outputs for GTQ Mini at various TAS.

TAS	Pitch	d_t	I_d	Endurance	Range
m/s	°	%	A	min	km
0	0	39	4.24	7.18	0
1.52	9.1	40	4.37	6.99	0.64
3.05	14.6	41.3	4.58	6.7	1.22
4.57	18.6	43.4	4.97	6.22	1.7
6.1	22.1	46.2	5.54	5.63	2.06
7.62	25.6	49.7	6.28	5.01	2.29
9.14	28.1	53.4	7.12	4.46	2.45
10.67	30.6	57.7	8.13	3.94	2.52
12.19	33.1	62.7	9.22	3.5	2.56
13.72	35.1	68	10.33	3.14	2.58
15.24	37.1	73.7	11.21	2.9	2.65

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